Time Domain and Frequency Domain Analysis

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Introduction

What is TSV?

- Through-Silicon Via (TSV) is a vertical electrical interconnect that passes through a silicon substrate to provide communication between stacked dies in 3D Integrated Circuits .

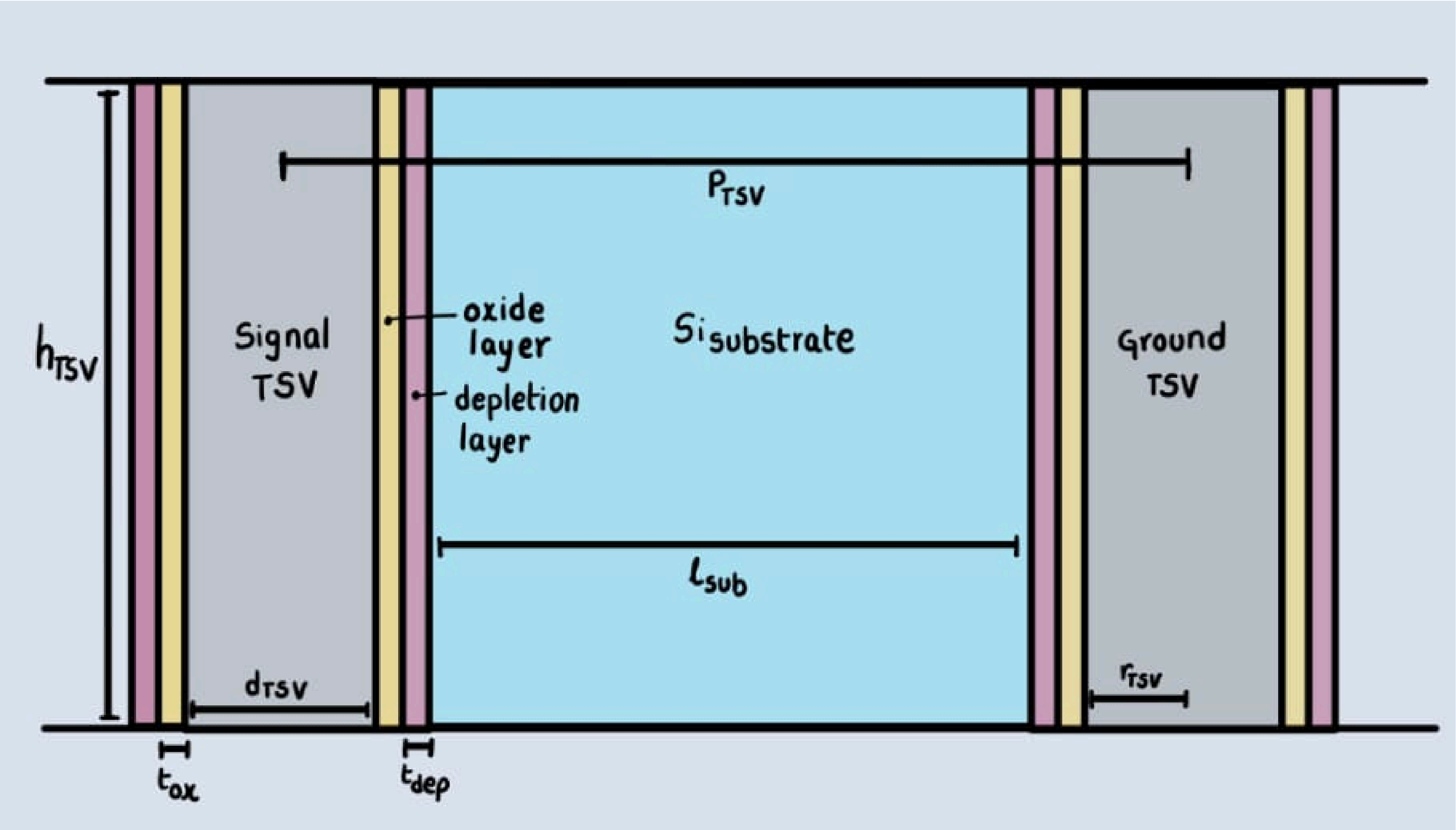

Fig.1 Structure of a signal TSV (Aggressor) and a ground TSV (Victim)

Introduction

Types of Defects During Fabrication of TSV

- **Void Defect:**

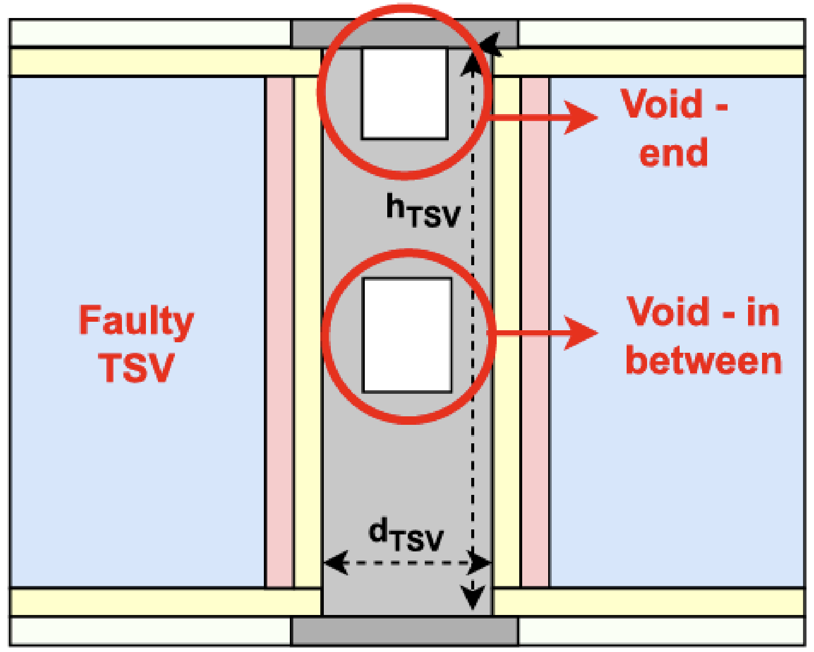

Fig.2.a. Void Defect[(3)]

- **Delamination Defect:**

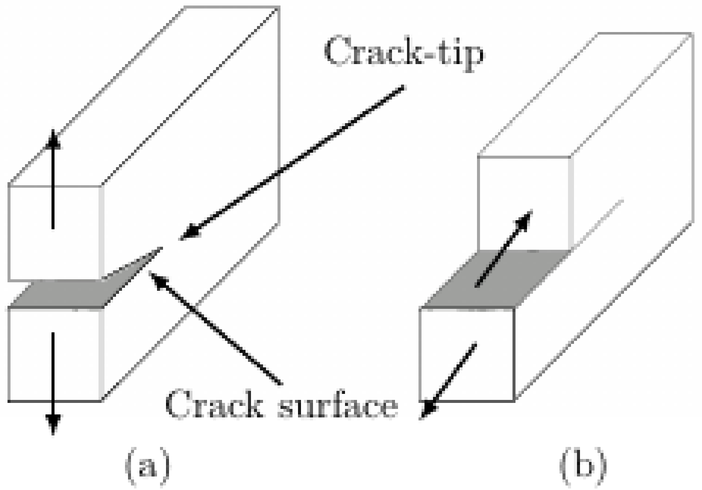

Fig.2.b. Delamination defect

- **Pinhole Defect:**

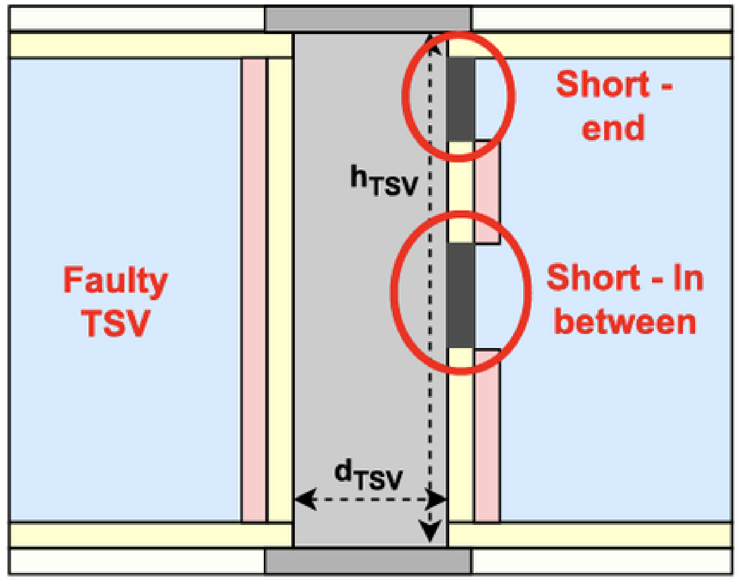

Fig.2.c. PinHole Defect[(3)]

- **Misalignment:**

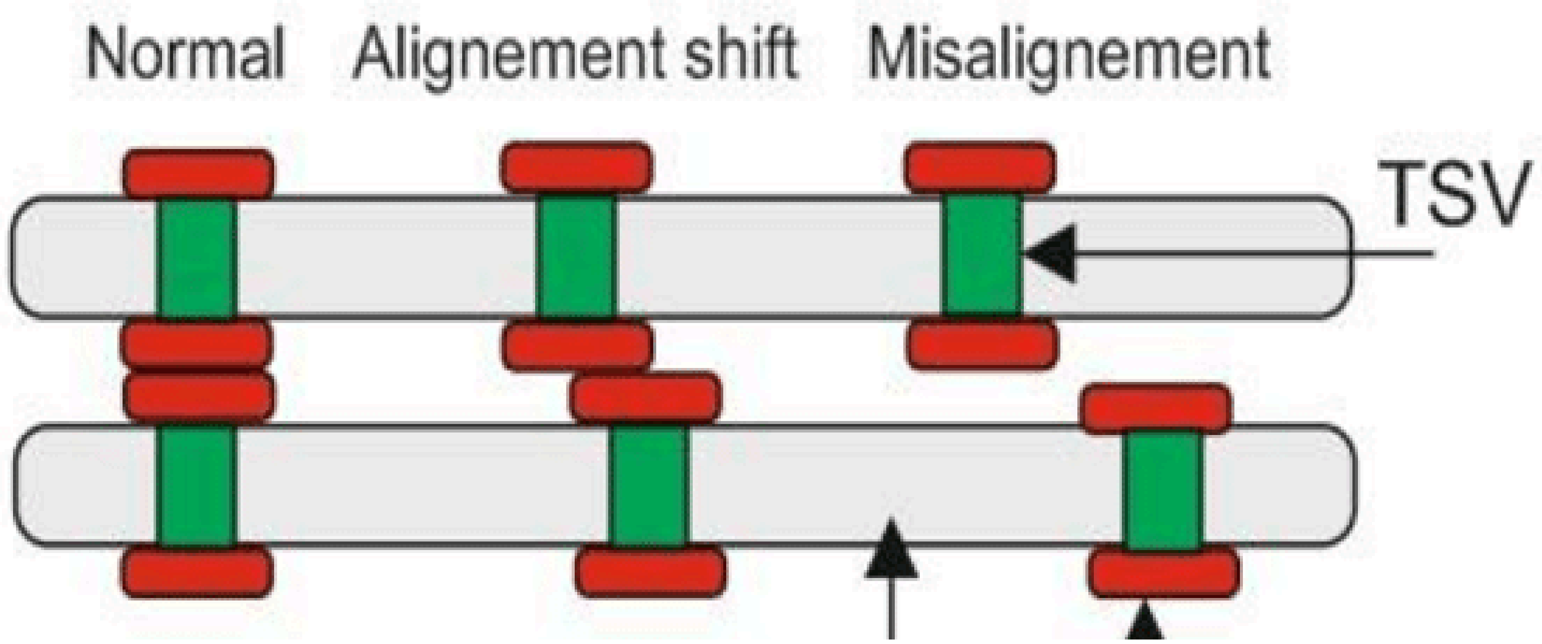

Fig.2.d. Misalignment Defect

- **Electro-Migration:**

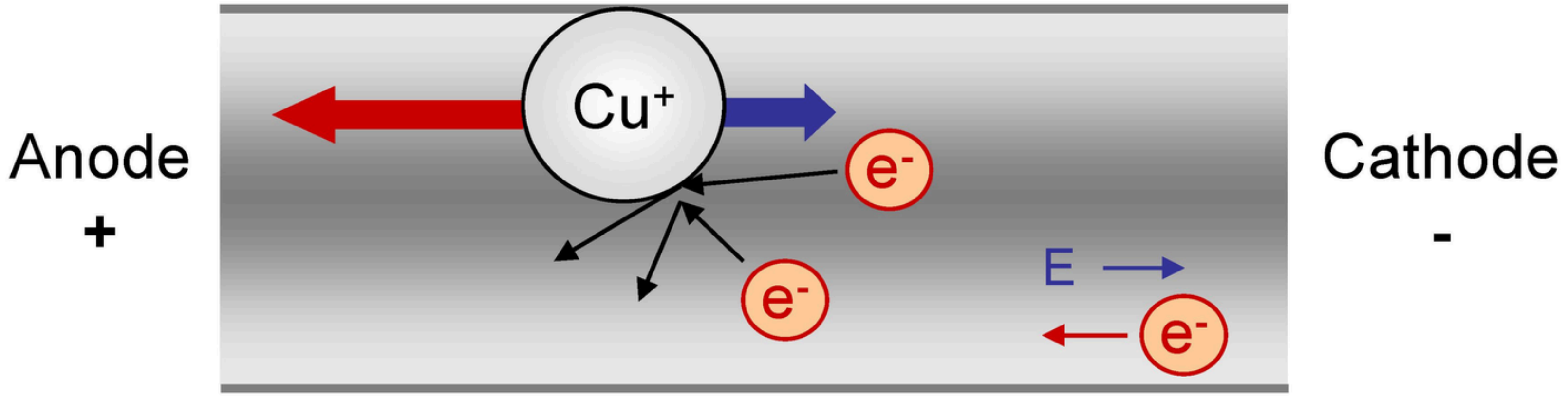

Fig.2.e. ElectroMigration

Focusing PinHole Defect

- Pinhole defects are one of the most common dielectric liner defects in TSVs.
- They create an unintended leakage path between the TSV and silicon substrate.
- They increase leakage current and leakage power.
- They degrade signal integrity by affecting transmission characteristics.
- They can increase noise, crosstalk, and propagation delay.
- They reduce the overall reliability of 3D ICs.

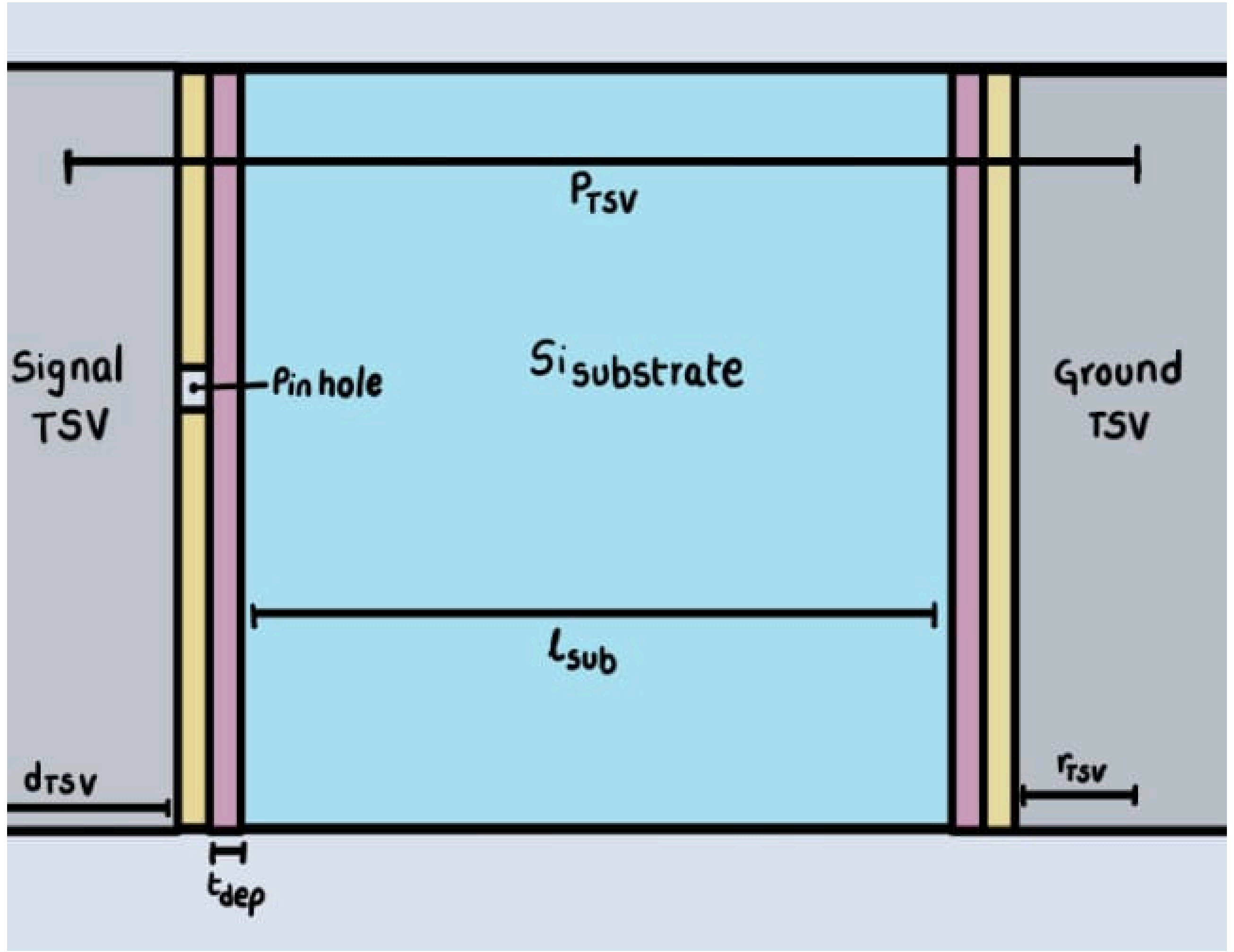

Fig.3. PinHole Defect

Physical Structure and Equivalent Electrical Structure of TSV

Defect-Free TSV Model

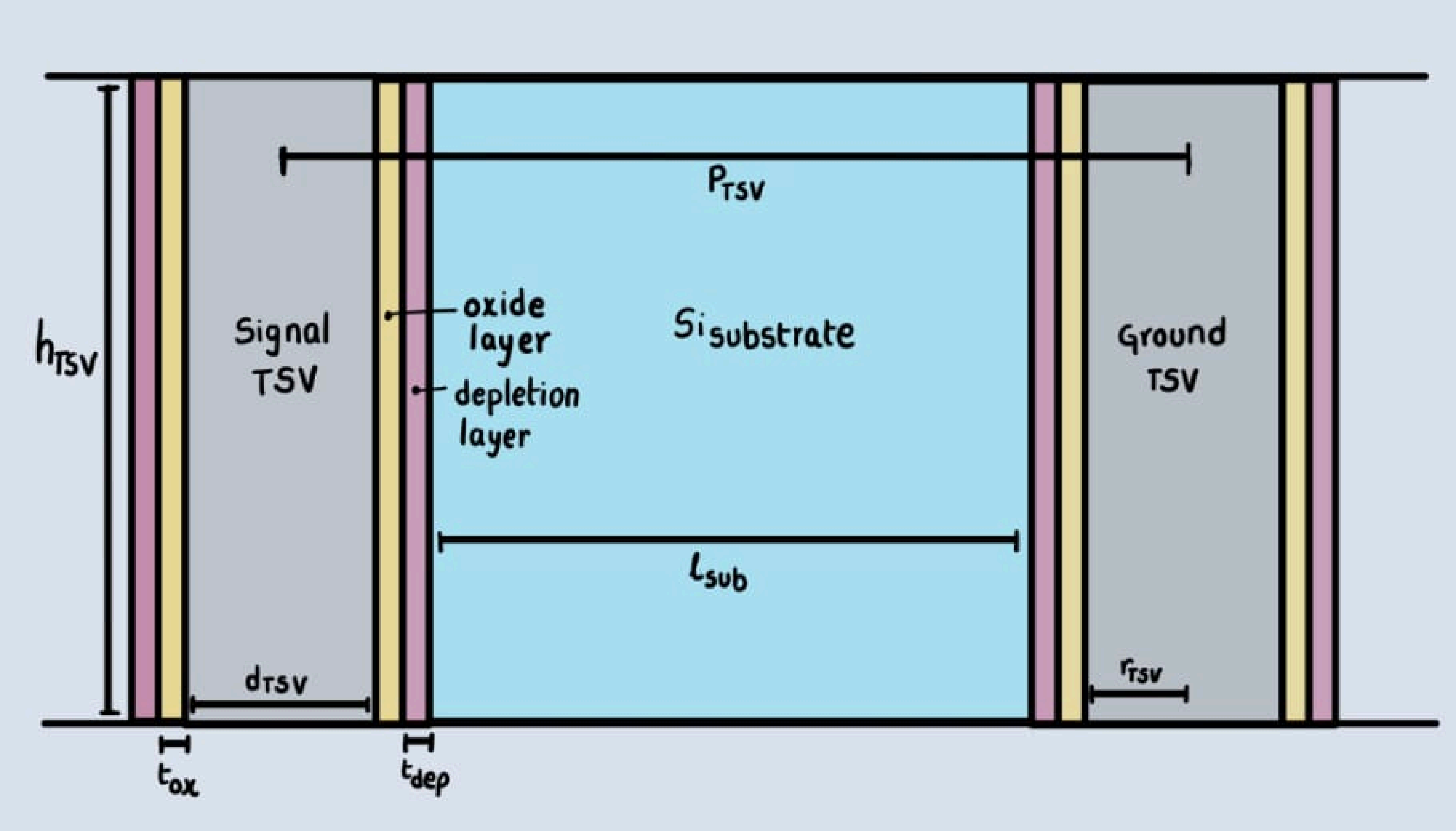

Fig.6.a Defect free TSV Physical Structure

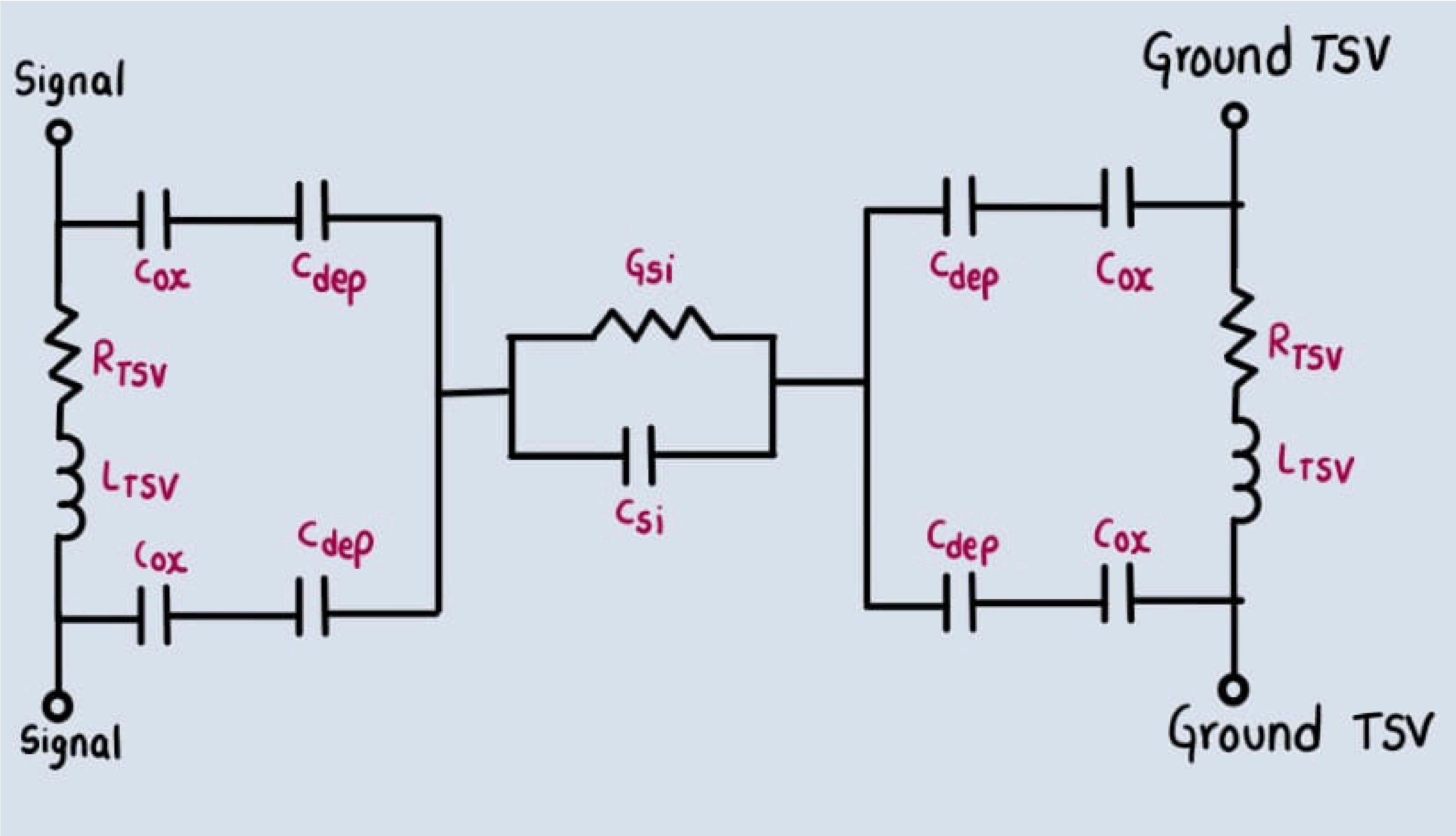

Fig.6.b Defect Free TSV Equivalent Electrical Model

Physical Structure and Equivalent Electrical Structure of TSV

Pinhole Defected Coupled via (Rsht and Lsht)

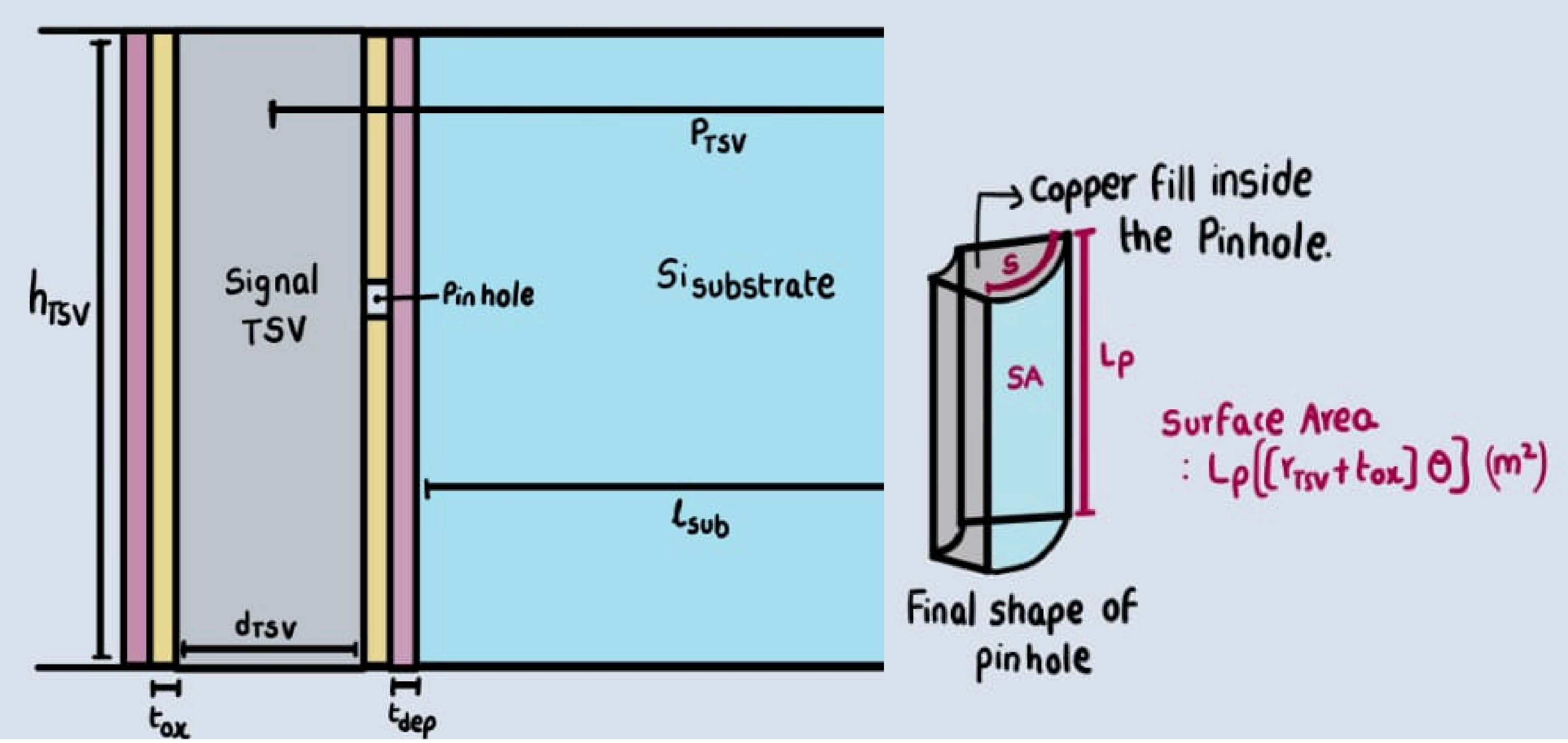

Fig.7.a Defected TSV Physical Structure

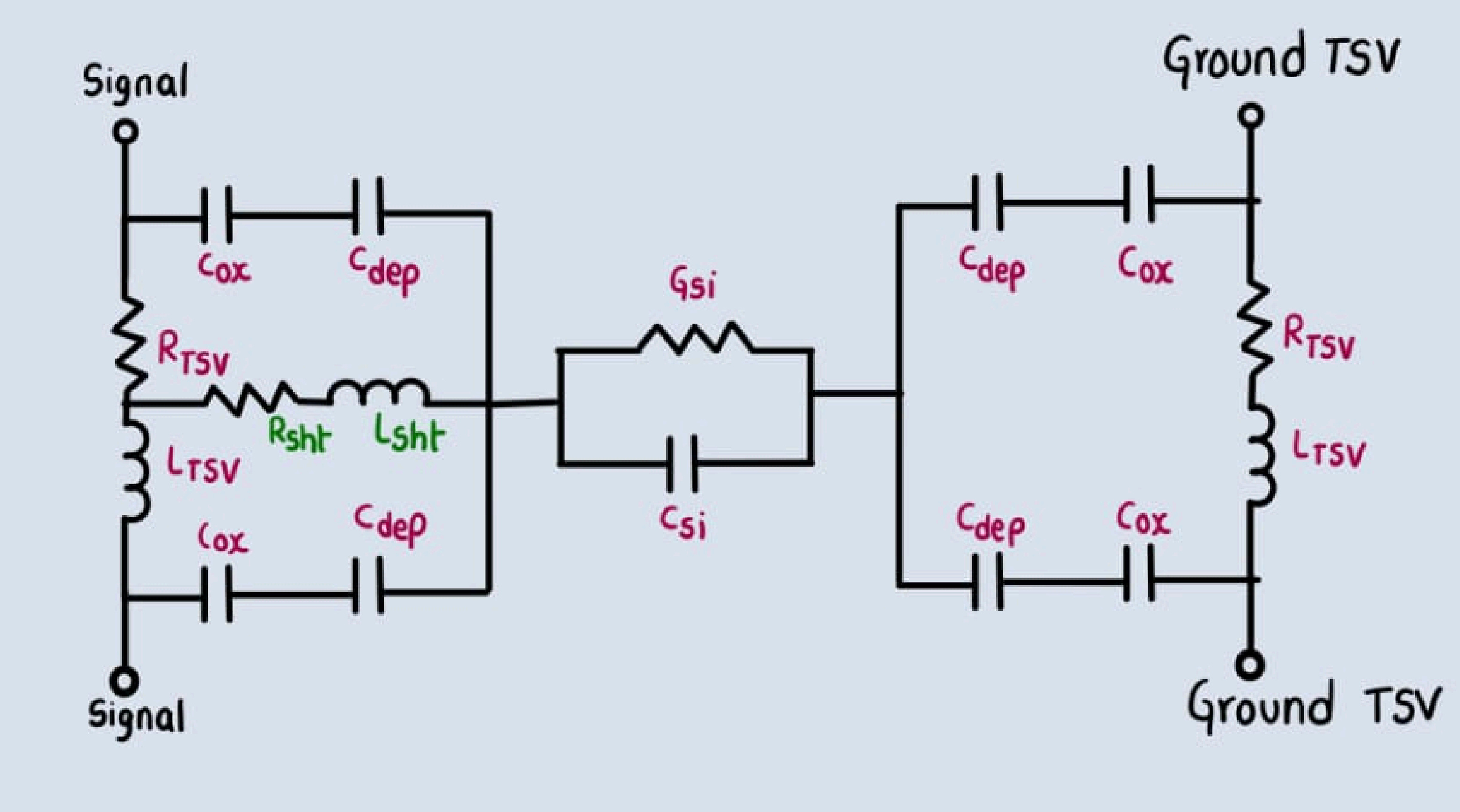

Fig.7.b Equivalent Electrical Circuit of Defected TSV

Physical Structure and Equivalent Electrical Structure of TSV

Pinhole Defected Coupled Via (Cleak and Gleak)

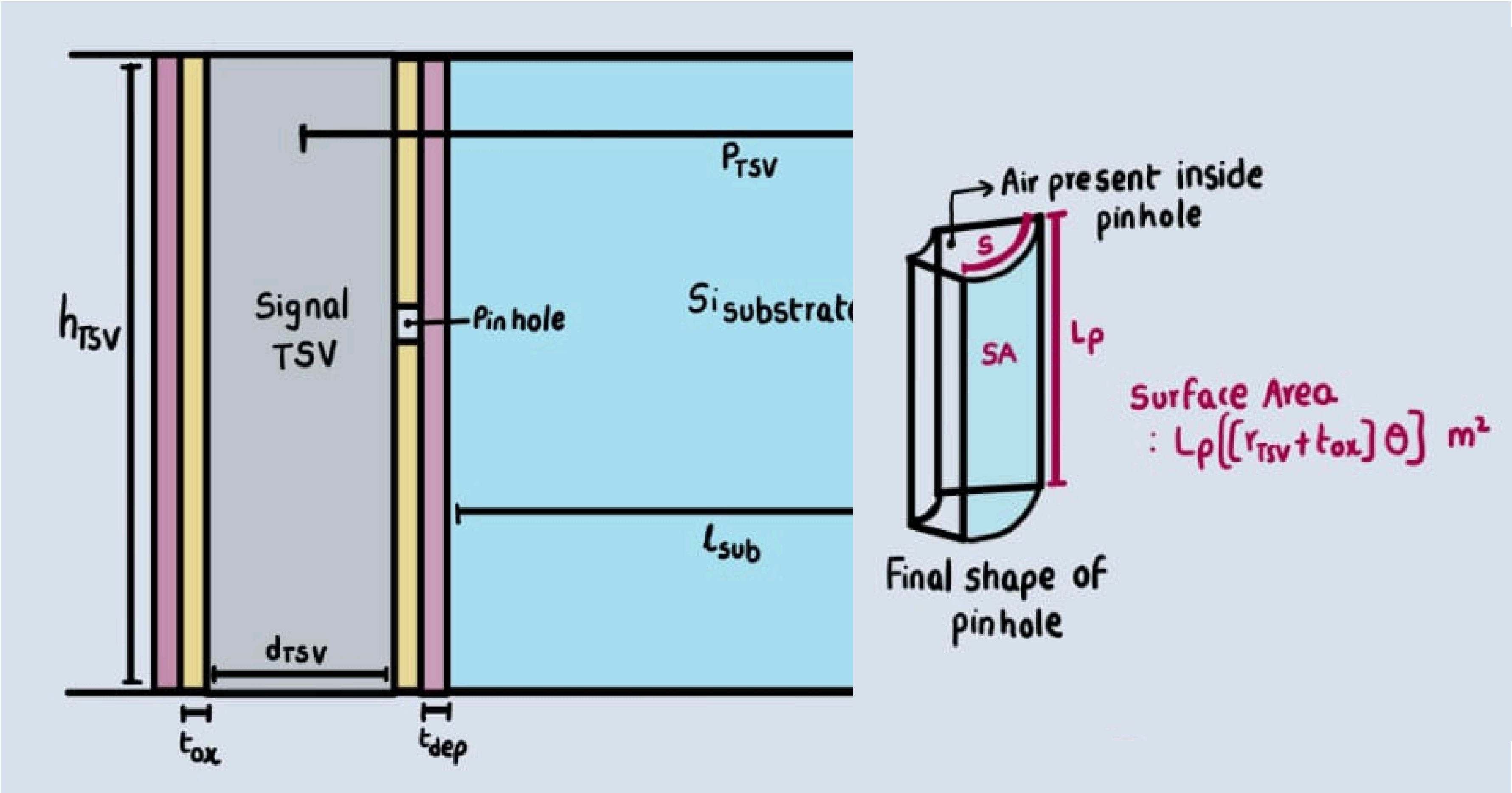

Fig.8.a TSV Defected Structure with Pinhole Cylindrical view

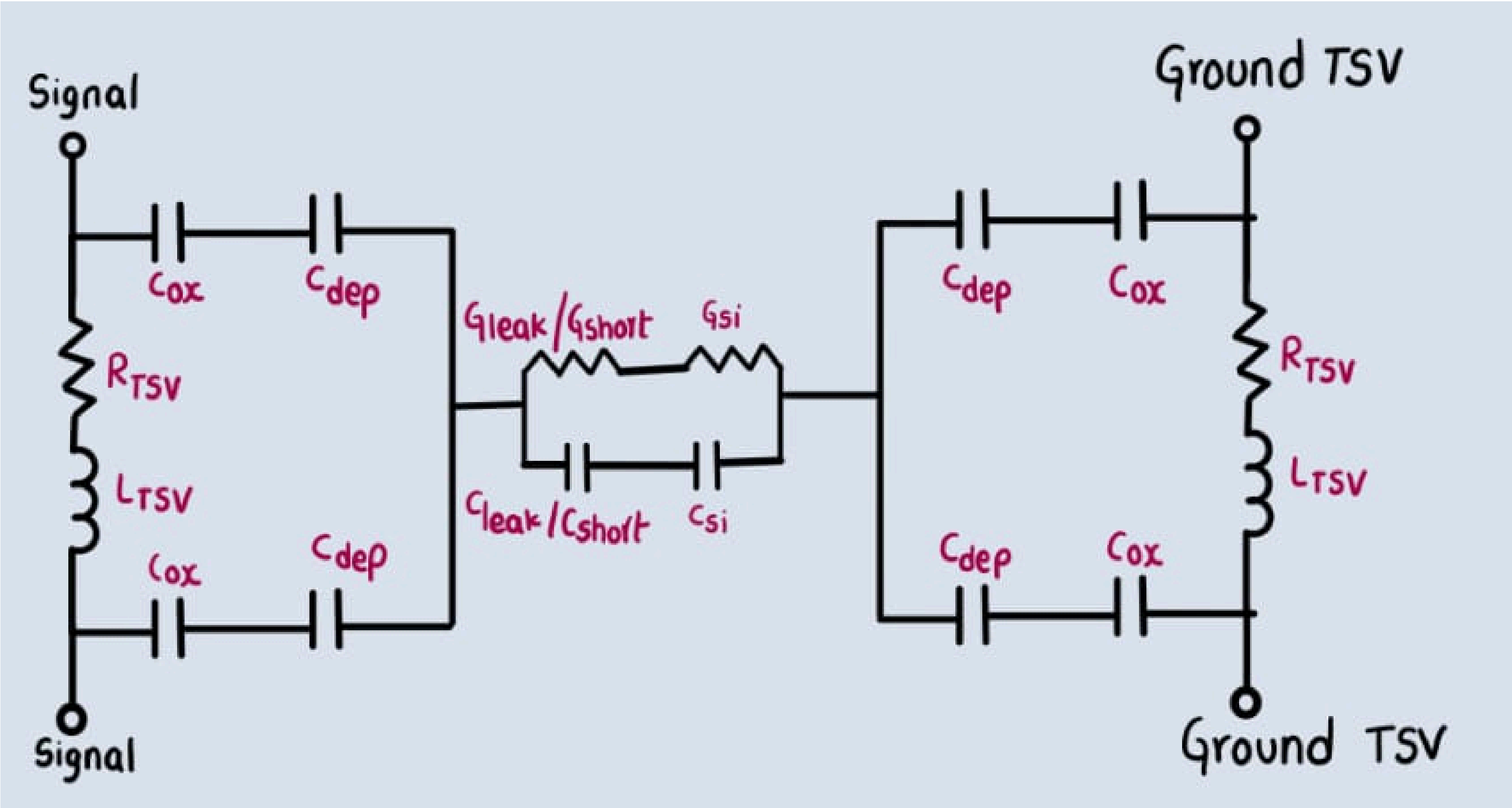

Fig.8.b Equivalent Electrical Circuit of Defected TSV

Equations

Defect-Free TSV Model

$$R_{TSV} = \sqrt{Rdc^2 + Rac^2}\,[ohms] \qquad \text{...Eq1}$$

Where

$$Rdc = \frac{\rho_{TSV} h_{TSV}}{\pi (r_{TSV})^2}[ohms] \qquad \text{...Eq2}$$

$$R_{ac} = \frac{\rho_{TSV} h_{TSV}}{2\pi r_{TSV}\delta_{sd} - \pi {\delta_{sd}}^2}[ohms] \qquad \text{...Eq3}$$

$$\delta_{sd} = \frac{1}{\sqrt{\pi f \mu_{TSV}\sigma_{TSV}}}[m] \qquad \text{...Eq4}$$

$$L_{TSV} = \left[\frac{1}{2}\frac{\mu_{TSV} h_{TSV}}{2\pi}\left[\ln\frac{\rho_{TSV}}{r_{TSV}}\right]\right][H] \qquad \text{...Eq5}$$

$$G_{si} = \frac{\pi \sigma_{si} h_{TSV}}{\cosh^{-1}\left[\frac{\rho_{TSV}}{d_{TSV}}\right]}[s] \qquad \text{...Eq6}$$

$$C_{ox} = \frac{2\pi\epsilon_{ox}\left[h_{TSV}\right]}{ln\left[\frac{r_{TSV}+tox+tdep}{r_{TSV}+tox}\right]}[F] \qquad \text{...Eq8}$$

$$C_{dep} = \frac{2\pi\epsilon_{si}\left[h_{TSV}\right]}{ln\left[\frac{r_{TSV}+tox+tdep}{r_{TSV}+tox}\right]}[F] \qquad \text{...Eq9}$$

$$C_{si} = \frac{\pi\epsilon_{si}\left[h_{TSV}\right]}{cosh^{-1}\left[\frac{\rho_{TSV}}{d_{TSV}}\right]}[F] \qquad \text{...Eq10}$$

$$Cins = \frac{CdepCox}{Cdep + Cox}[F] \qquad \text{...Eq11}$$

Equations

Pinhole Defected Coupled via (Rsht and Lsht)

- Rtsv, Ltsv, Gsi, Csi, Cox, Cdep equations are the same as Defect Free model
- Eq12, Eq13 are Reference Model Equations
- Eq14, Eq15 are Proposed Model Equations

$$Rsht = \frac{\rho_T tox}{Lp^2}[ohms]$$ ...Eq12

$$Lsht = \frac{\mu_{TSV} t_{ox}}{2\pi}\left[arccosh\left[\frac{tox}{\sqrt{Lp^2 + tox^2}}\right] - \sqrt{2tox^2 + Lp^2} + \frac{tox}{4} + \sqrt{tox^2 + Lp^2}\right][H]$$ ...Eq13

$$Rsht = \frac{\rho_T tox}{Lp\,[r_{TSV} + tox]\theta}[ohms]$$ ...Eq14

$$Lsht = \frac{\mu_{TSV} t_{ox}}{2\pi}\left[arccosh\left[\frac{S}{\sqrt{Lp^2 + S^2}}\right] - \sqrt{S^2 + tox^2 + Lp^2} + \frac{S}{4} + \sqrt{S^2 + Lp^2}\right][H]$$ ...Eq15

Equations

Pinhole Defected Coupled Via (Cleak and Gleak)

- Rtsv, Ltsv, Cox, Cdep equations are the same as Defect Free model , Gsi, Csi will change due to the leakage capacitance and conductance.
- Eq18, Eq20 are Reference Model Equations
- Eq21, Eq22 are Proposed Model Equations

$$J = \frac{\theta Lp}{2\pi r_{TSV}} \quad \text{...Eq19}$$

$$G_{si} = \frac{\pi\sigma_{si}\left[h_{TSV} - Lp\right]}{\cosh^{-1}\left[\frac{\rho_{TSV}}{d_{TSV}}\right]}[s] \quad \text{...Eq16}$$

$$Gleak = \frac{2\pi\sigma_{ox}h_{TSV} - 4\pi^2\epsilon_{air}Jfr_{TSV}}{ln\left[\frac{r_{TSV}+tox}{r_{TSV}}\right]}[S] \quad \text{...Eq20}$$

$$C_{si} = \frac{\pi\epsilon_{si}\left[h_{TSV} - Lp\right]}{cosh^{-1}\left[\frac{\rho_{TSV}}{d_{TSV}}\right]}[F] \quad \text{...Eq17}$$

$$Cshort = \frac{[2\pi - \theta]\epsilon_{ox}Lp + \theta\epsilon_{air}Lp]}{ln\left[\frac{r_{TSV}+tox}{r_{TSV}}\right]}[F] \quad \text{...Eq21}$$

$$Cleak = \frac{2\pi\epsilon_{ox}h_{TSV} - 2\pi\epsilon_{air}Jr_{TSV}}{ln\left[\frac{r_{TSV}+tox}{r_{TSV}}\right]}[F] \quad \text{...Eq18}$$

$$Gshort = \frac{[2\pi - \theta]\sigma_{ox}Lp + \theta\epsilon_{air}Lpf}{ln\left[\frac{r_{TSV}+tox}{r_{TSV}}\right]}[S] \quad \text{...Eq22}$$

Objective

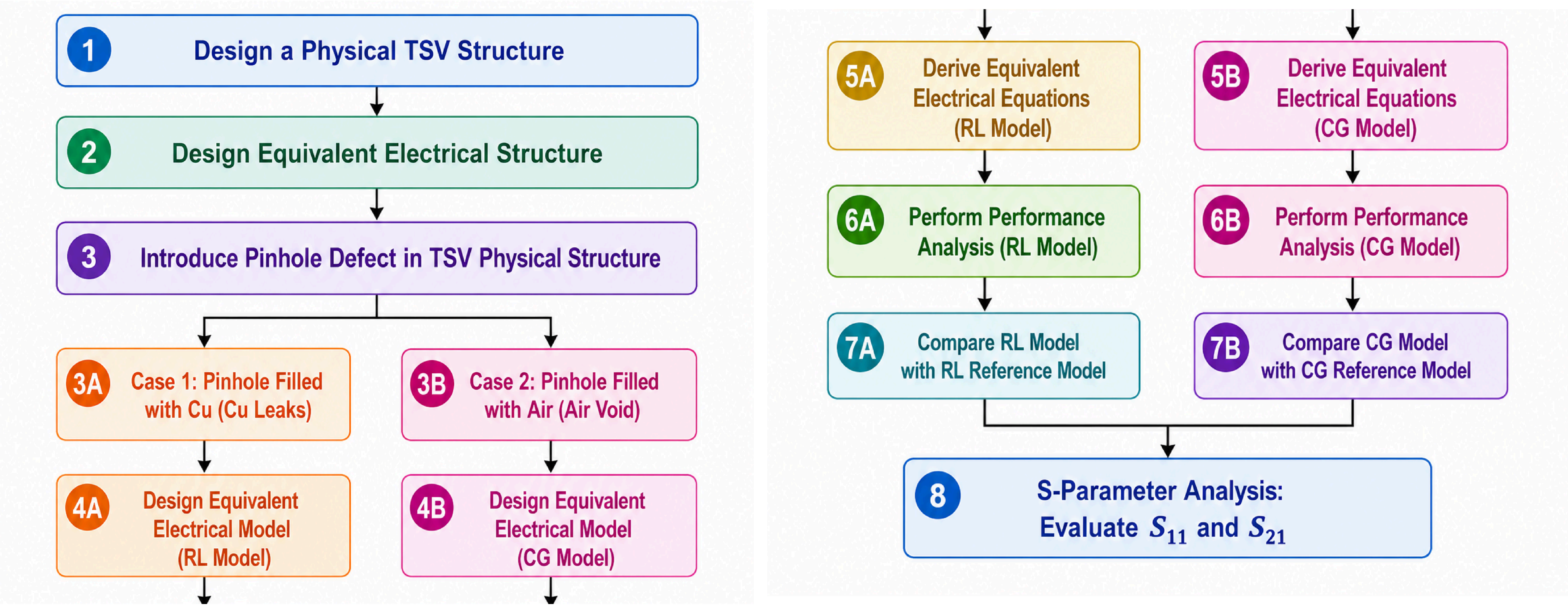

Fig. 5. Objective Flow Chart

TSV DESIGN SPECIFICATIONS

Table.1.a Parameters of the TSV model[5]

Symbol	Meaning	Value	Units
pTSV	Pitch between two TSV's[=4rTSV]	10	µm
dTSV	Diameter of TSV	4.94	µm
rTSV	Radius of TSV	2.47	µm
hTSV	height of TSV	15	µm
tox	Thickness of oxide layer	0.13	µm
tdep	Thickness of Deplection layer	0.87	µm
l substrate	length of the Substrate	3.06	µm
σ si	Conductivity of Silicon Substrate	10	S/m
σ cu	Conductivity of copper	59000000	S/m
ρTSV	Resistivity of TSV	0.000000000168	Ω·m
εr,si	Relative Permittivity of Silicon	11.9	No Unit
εr,ox	Relative permittivity of oxide layer	4	No Unit
µr,TSV	Relative permeability of TSV	1	No Unit
δox	loss tangent of oxide layer	0	No Unit
δsi	loss tangent of Silicon	0	No Unit

TSV DESIGN SPECIFICATIONS

Fig. 1.b Parameters of the TSV model[5]

Symbol	Meaning	Value	units
J	Leakage Factor	40%	No unit
θ	Angle of the Sector [Pinhole]	π/2	Radians
Lp	Length of the Sector [Pinhole]	4	μm
T	The temperature at which you want to find the resistivity	298	K
T0	Reference temperature	273	K
α	Temperature coefficient of resistivity	0.039	K^{-1}
Lp	Length of the sector [pinhole]	4	μm
ρ0	Resistivity at the reference temperature	0.0000000168	Ω·m
ρT	Resistivity of the material at temperature	0.0000000332	Ω·m
θ	Angle of the Sector [pinhole]	π/2	Radians

Pinhole Defect 2 Cases

Case 01:

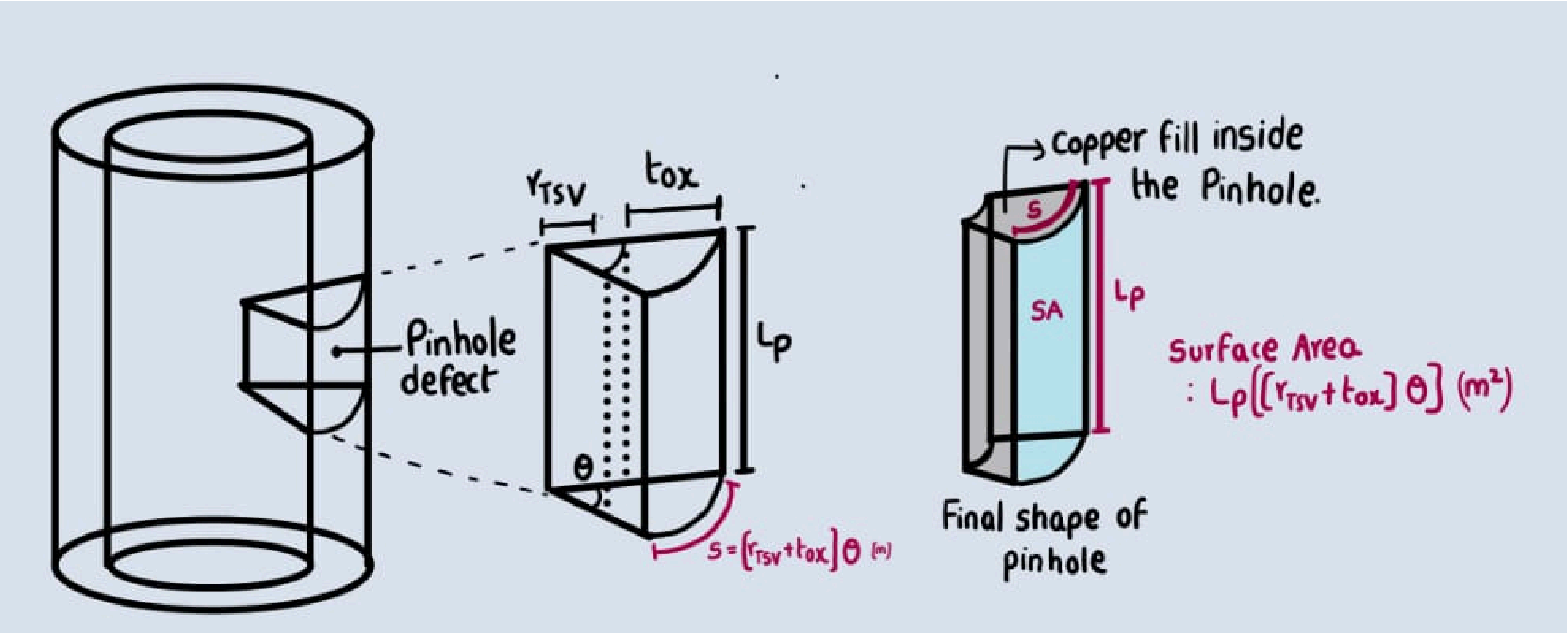

Fig.4.a Pinhole Filled with leaked Copper

- Occurs mainly during: Copper seed deposition and copper electroplating stages.

Case 02:

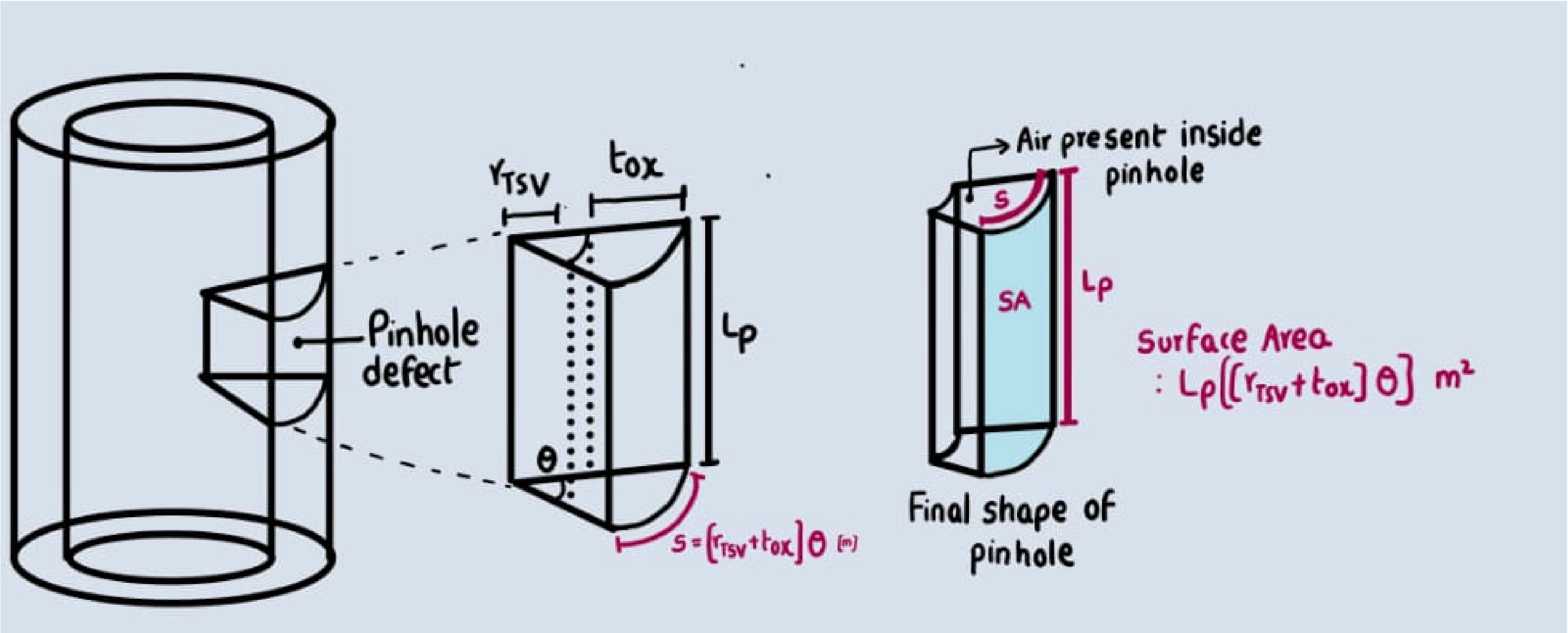

Fig.4.b Pinhole Filled with Air Trapped

- Occurs mainly during: Oxide liner deposition

Parasitic Parameters

Defect-Free TSV Model

Table. 2.a Parasitic Parameters values of Defect Free Model

Full form	Parasitic Parameter	Defect-Free
Through-Silicon Via Resistance	RTSV [Ω]	82.8m
Through-Silicon Via Inductance	LTSV [H]	2.09p
Insulation Capacitance	CIns [F]	11.55f
Oxide Capacitance	COX [F]	65.07f
Depletion Capacitance	CDep [F]	35.84f
Substrate Capacitance	CSub [F]	3.73f
Substrate Conductance	GSub [S]	354u

Parasitic Parameters

Pinhole Defected Coupled via Models

Table. 2.b Parasitic Parameters values of Pinhole Defected Coupled via Models

Symbol Meaning	Symbol	Defected TSV Pinhole [Proposed Model]	Defected TSV Pinhole [Reference Model]
Pinhole Defect Resistance	RSht [Ω]	0.133m	0.136m
Pinhole Defect Inductance	LSht [H]	2.05E-20	1.901E-20
Pinhole Defect Capacitance	CSht [F]	14.09f	64f
Pinhole Defect Conductance	GSht [S]	108u	214u

Performance analysis Arrangement Setup :

Table.3 Performance analysis setup for Pinhole Defected Coupled via Models

Symbol	Meaning	Value	units
Vin	Signal Voltage	0.8	V
CL	Load Capacitance	1f	F
DT	Driver Technology	22n	m

Performance Analysis

Noise

Fig.9 Noise analysis of the proposed model and Reference models

Crosstalk-Delay

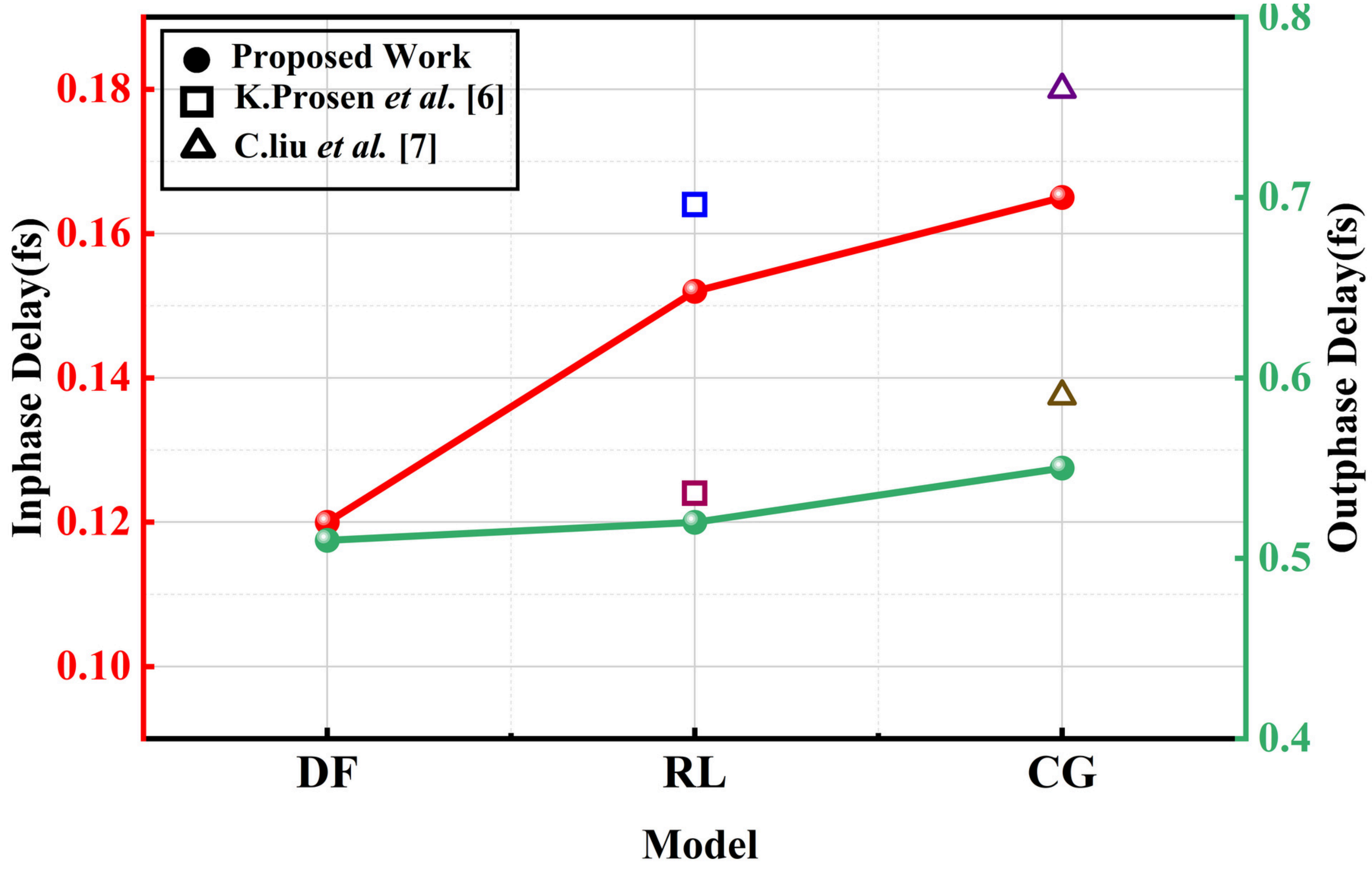

Fig.10 Crosstalk delay analysis of the proposed model and Reference models

Performance Analysis

Leakage Current and Power

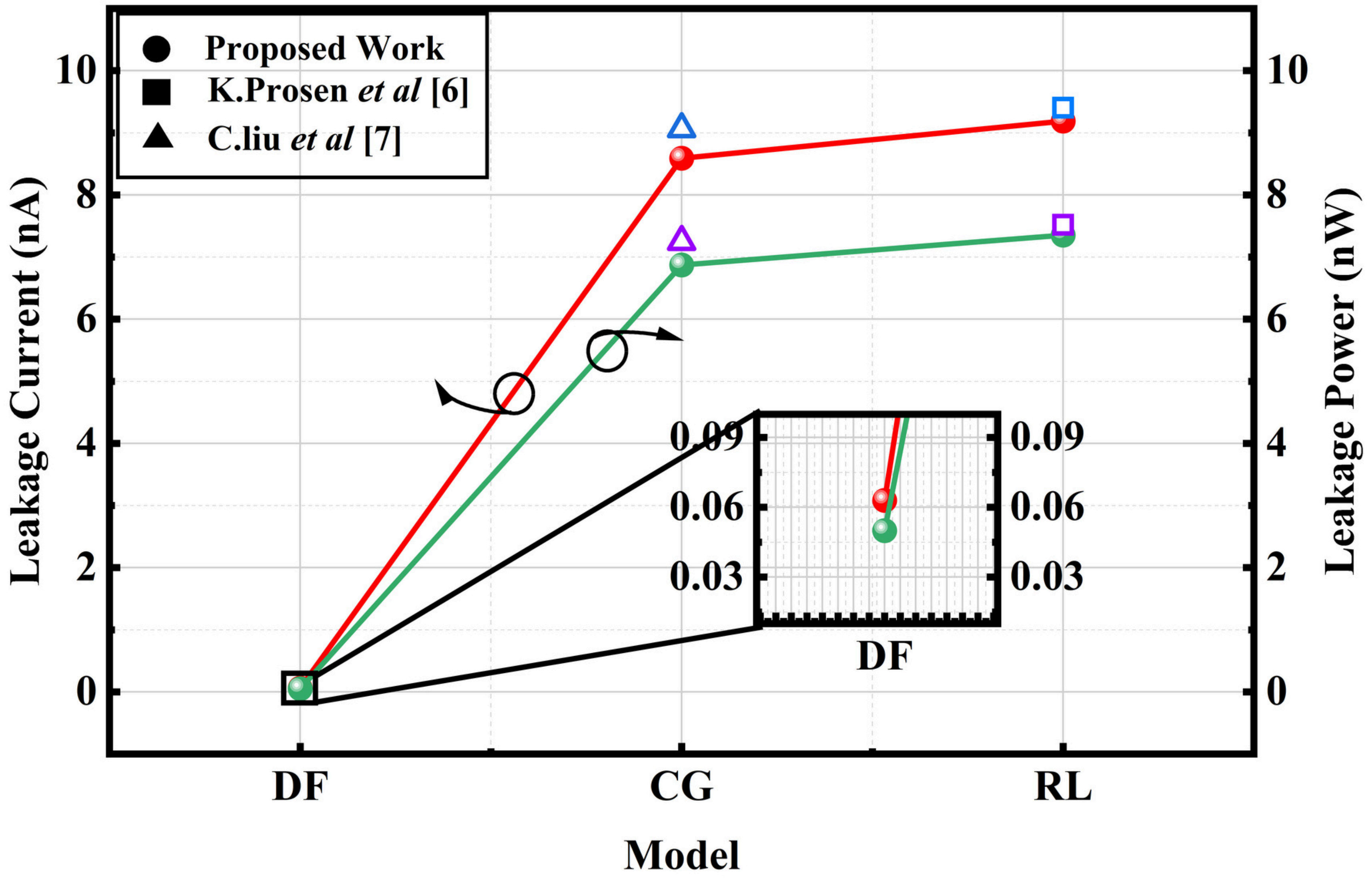

Fig.11 Leakage Current and Power analysis of the proposed model and Reference models

Propagational Delay

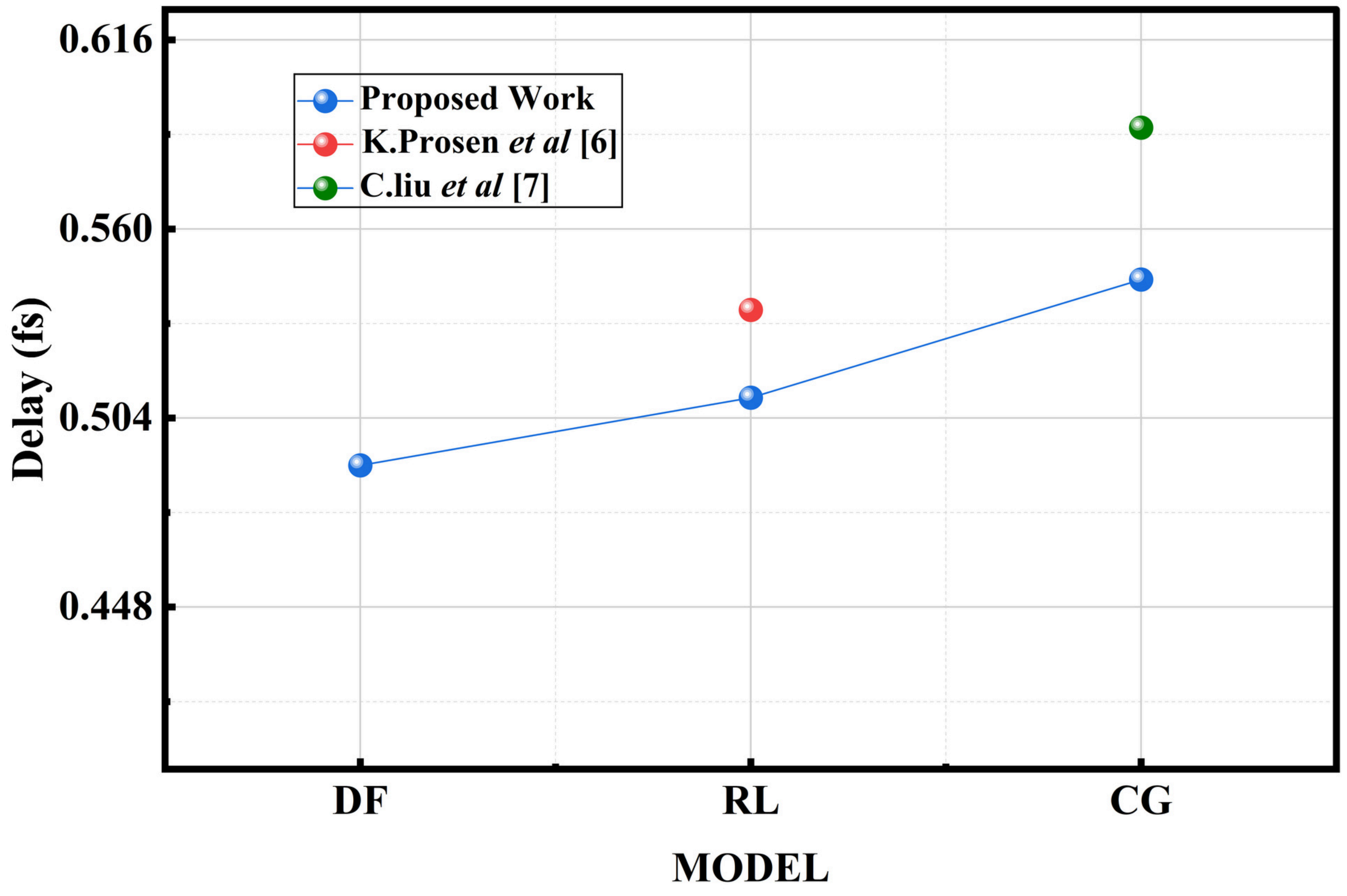

Fig.12 Propagation Delay analysis of the proposed models and Reference models

S Parameters

S_{11} Parameter

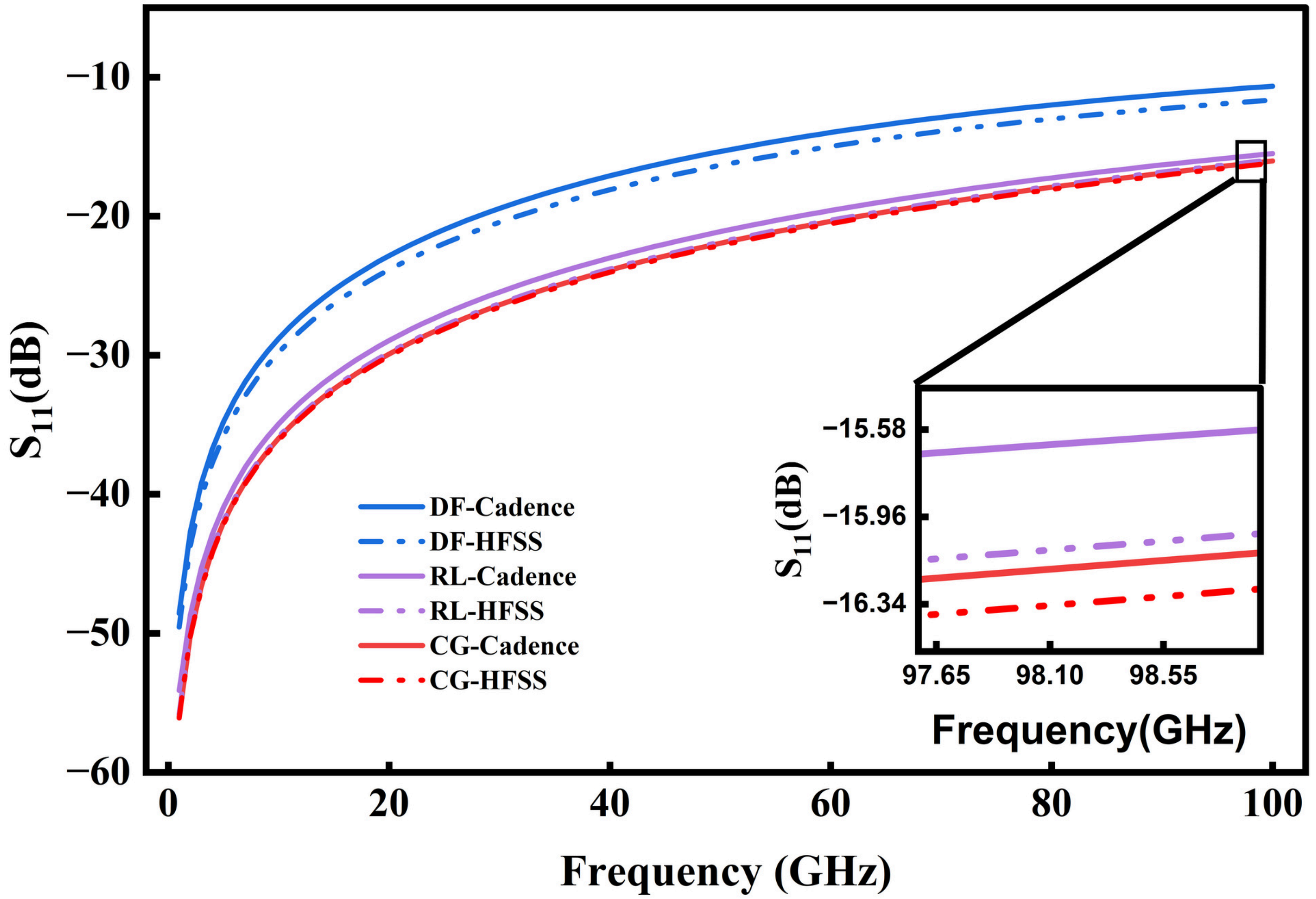

Fig. 13. Reflection coefficient comparison of the Proposed Model with the Reference Model

S_{21} Parameter

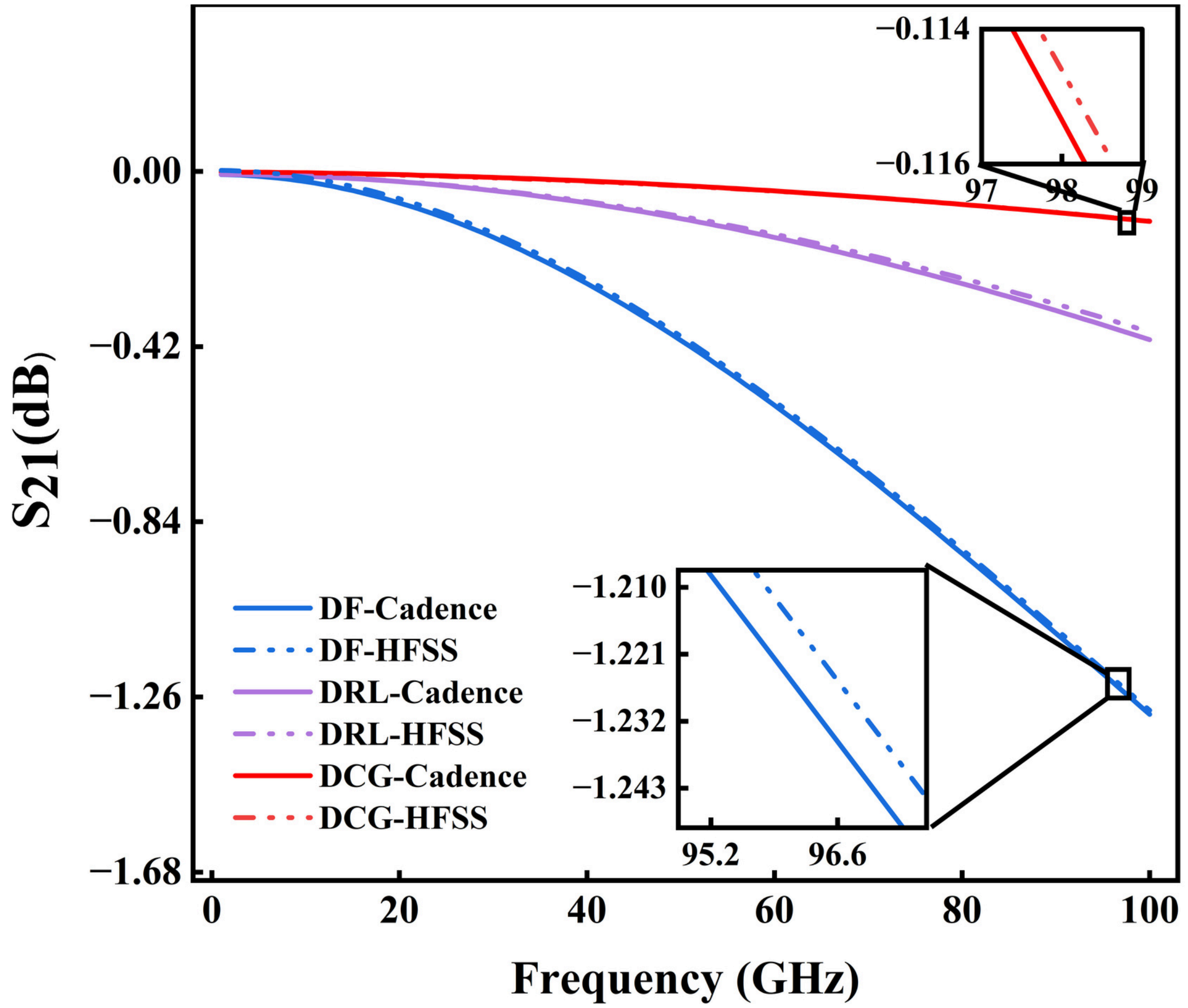

Fig. 14. Transmission coefficient comparison of the Proposed Model with the Reference Model

References

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